

Tianzhe Chu

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Google Scholar · GitHub

RESEARCH HIGHLIGHTS

Benchmarking Dexterous Embodied System

A real-world benchmark probing whether embodied agents can perceive the table, track the evolving game state across turns, and act on it through dexterous manipulation, with Texas Hold'em as a rich physical testbed. As co-leader, I led the embodied agent design and the perception benchmark measuring how reliably agents recover world state from raw observation.

[\[paper\]](#) [\[website\]](#) [\[code\]](#)

SFT Memorizes, RL Generalizes

Under equal training compute, RL with verified rewards generalizes to unseen textual and visual variants while SFT memorizes the training distribution, though SFT remains a necessary format-stabilization step before RL. Further shows that scaling up verification steps during RL training boosts its generalization.

[\[paper\]](#) [\[website\]](#) [\[code\]](#)

EXPERIENCE

Advanced Machine Intelligence

Research Intern, World Model

Singapore

Starting Sep 2026

Hyperbolic

Research Intern, Agent and AI4Math

San Francisco, CA, US

Dec 2025 – Feb 2026

University of California, Berkeley

Visiting Undergraduate, EECS & BAIR

Berkeley, CA, US

Aug 2022 – May 2023

EDUCATION

The University of Hong Kong

Ph.D. candidate at IDS advised by Prof. Yi Ma

Hong Kong SAR

Sep 2024 – Present

ShanghaiTech University

B.Eng in Computer Science and Technology (Honor)

Shanghai, China

Sep 2020 – Jun 2024

PUBLICATIONS

(* denotes equal contribution)

SFT Memorizes, RL Generalizes: A Comprehensive Study of Foundation Model Post-training

Tianzhe Chu*, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, Yi Ma
ICML 2025 [\[paper\]](#) [\[website\]](#) [\[code\]](#)

Pointer-CAD: Unifying B-Rep and Command Sequences via Pointer-based Edges & Faces Selection

Dacheng Qi, Chenyu Wang, Jingwei Xu, Tianzhe Chu, Zibo Zhao, Wen Liu, Wenrui Ding, Yi Ma, Shenghua Gao
CVPR 2026 [\[paper\]](#)

Seeing from Another Perspective: Evaluating Multi-View Understanding in MLLMs

Chun-Hsiao Yeh*, Chenyu Wang*, Shengbang Tong, Ta-Ying Cheng, Ruoyu Wang, Tianzhe Chu, Yuexiang Zhai, Yubei Chen, Shenghua Gao, Yi Ma
AAAI 2026 [\[paper\]](#) [\[website\]](#) [\[code\]](#)

White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?

Yaodong Yu, Sam Buchanan, Druv Pai, Tianzhe Chu, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haefele, Yi Ma
JMLR [\[paper\]](#)

Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models

Tianzhe Chu*, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma

ICLR 2024 [[paper](#)] [[code](#)]

Emergence of Segmentation with Minimalistic White-Box Transformers

Yaodong Yu*, Tianzhe Chu*, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma

CPAL 2024 (**Oral**), NeurIPS 2023 XAI Workshop (**Oral**) [[paper](#)] [[website](#)] [[code](#)]

White-Box Transformers via Sparse Rate Reduction

Yaodong Yu, Sam Buchanan, Druv Pai, Tianzhe Chu, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma

NeurIPS 2023 [[paper](#)] [[website](#)] [[code](#)]

DexHoldem: Playing Texas Hold'em with Dexterous Embodied System

Feng Chen*, Tianzhe Chu*, Li Sun*, Pei Zhou*, Zhuxiu Xu, Shenghua Gao, Yuexiang Zhai, Yanchao Yang, Yi Ma

Preprint [[paper](#)] [[website](#)] [[code](#)]

GenDexHand: Generative Simulation for Dexterous Hands

Feng Chen*, Zhuxiu Xu*, Tianzhe Chu, Xunzhe Zhou, Li Sun, Zewen Wu, Shenghua Gao, Zhongyu Li, Yanchao Yang, Yi Ma

Preprint [[paper](#)]

GAUSS Eval: Human-LLM Judge Consistency Analysis

Tianzhe Chu*, Jiaxin Zhang*, Zhenyu Liao, Qiuyu Ren, Tahsin Saffat, Zitong Yang, Yi Ma, Yue Zhang*

Blog Post [[website](#)]

OPEN SOURCE

Deep Representation Learning Book — Core Contributor [[code](#)]

Open-source textbook on the principles of deep representation learning.

CRATE — Core Contributor [[code](#)]

White-box transformer architecture derived from sparse rate reduction.

SELECTED AWARDS

Outstanding Graduate

ShanghaiTech University

Shanghai, China

Jul 2024

Merit Student

ShanghaiTech University

Shanghai, China

Dec 2023

INVITED TALKS

- Localize the Credits of Reasoning in Roaring 2025: LLM Reasoning Paper Potpourri @ ALICE, Aug. 2025
- SFT Memorizes, RL Generalizes @ BAAI, Jun. 2025
- SFT Memorizes, RL Generalizes @ Cathy Wu's lab, MIT, Apr. 2025
- White-Box Transformers @ SVIP Lab, ShanghaiTech, Sep. 2023
- Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, Jan. 2024

SERVICE

Head Teaching Assistant — HKU DATA 8014

Reviewer — ICLR 2025/2026, ICML 2025, NeurIPS 2025, CVPR 2026